No.	Status	Date
LNDE-RPT-20160118-004-DS-4-RA-102	Issued	2016-02-12

UT Thickness Data Sheet - 4-RA-102

Scope and Responsibilities

This data sheet provides the ultrasonic thickness measurement results for line 4-RA-102 located within the ATU-100. The scope of work involves the recording of wall thickness to monitor the integrity of the rich amine service piping.

Marcus Chen, acting as the lead UT technician (SNT-TC-1A UT Level II) for Larkhall NDE Services Inc., performed the measurements and is responsible for the accuracy of the data recorded herein.

Technical Basis

All thickness measurements were conducted in accordance with the requirements of ASTM E797. The inspection was performed under the guidelines specified in QSCP-NDT-001 Rev 2, which includes requirements for insulation removal and reinstatement as part of the CUI program.

The data presented in this sheet supports the findings detailed in the full-unit survey report LNDE-RPT-20160118-004-R0.

Line Specifications

The subject line, 4-RA-102, is constructed from A106 Gr.B with a nominal outer diameter of 114.3 mm and a nominal wall thickness of 8.56 mm. The design pressure is 862 kPa(g) and the design temperature is 82 °C.

The minimum allowable wall thickness (t-min) for this line is 3.81 mm. Any reading below this threshold is subject to an NCR.

Inspection Interval and Classification

Line 4-RA-102 is classified as rich amine service, which mandates a 3-year thickness measurement interval.

The current survey is part of the Class 2 10-year cycle since 2006, while adhering to the specific 3-year requirement for rich amine circuits.

Thickness readings, measured 2016-02-12

Line 6-SG-101 | Sour gas inlet | A106 Gr.B | t_nom 7.11 mm | t-min 3.56 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
6-SG-101-01	7.11	3.56	2011-02-18	6.39	6.17	0.048	0.044	>50	2021-02-12	
6-SG-101-02	7.11	3.56	2011-02-18	6.14	5.76	0.065	0.076	28.8	2021-02-12	
6-SG-101-03	7.11	3.56	2011-02-18	6.17	5.87	0.061	0.060	38.1	2021-02-12	
6-SG-101-04	7.11	3.56	2011-02-18	6.05	5.58	0.076	0.094	21.4	2021-02-12	
6-SG-101-05	7.11	3.56	2011-02-18	5.87	5.38	0.088	0.098	18.5	2021-02-12	
6-SG-101-06	7.11	3.56	2011-02-18	6.27	5.88	0.060	0.078	29.6	2021-02-12	
6-SG-101-08	7.11	3.56	2011-02-18	6.33	6.00	0.066	0.066	36.8	2021-02-12	

Line 6-SG-102 | Sour gas inlet | A106 Gr.B | t_nom 7.11 mm | t-min 3.56 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
6-SG-102-01	7.11	3.56	2011-02-18	5.75	5.42	0.086	0.066	21.6	2021-02-12	
6-SG-102-02	7.11	3.56	2011-02-18	5.80	5.43	0.084	0.074	22.2	2021-02-12	
6-SG-102-03	7.11	3.56	2011-02-18	5.80	5.22	0.091	0.116	14.3	2021-02-12	
6-SG-102-04	7.11	3.56	2011-02-18	5.73	5.18	0.091	0.110	14.7	2021-02-12	
6-SG-102-05	7.11	3.56	2011-02-18	6.46	6.19	0.047	0.054	48.5	2021-02-12	
6-SG-102-06	7.11	3.56	2011-02-18	6.41	6.06	0.070	0.070	35.6	2021-02-12	

Line 8-TG-101 | Treated gas | A106 Gr.B | t_nom 8.18 mm | t-min 4.06 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
8-TG-101-01	8.18	4.06	2006-02-24	7.89	7.76	0.022	0.013	>50	2026-02-12	
8-TG-101-02	8.18	4.06	2006-02-24	7.75	7.46	0.037	0.029	>50	2026-02-12	
8-TG-101-03	8.18	4.06	2006-02-24	7.79	7.54	0.035	0.025	>50	2026-02-12	
8-TG-101-04	8.18	4.06	2006-02-24	7.76	7.51	0.030	0.025	>50	2026-02-12	
8-TG-101-05	8.18	4.06	2006-02-24	7.95	7.73	0.025	0.022	>50	2026-02-12	
8-TG-101-07	8.18	4.06	—	—	7.39	—	—	—	2026-02-12	

Line 4-LA-101 | Lean amine supply | A106 Gr.B | t_nom 6.02 mm | t-min 3.18 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
4-LA-101-01	6.02	3.18	2006-02-24	5.51	5.02	0.049	0.049	37.4	2026-02-12	
4-LA-101-02	6.02	3.18	2006-02-24	5.76	5.45	0.028	0.031	>50	2026-02-12	
4-LA-101-03	6.02	3.18	2006-02-24	5.76	5.45	0.026	0.031	>50	2026-02-12	
4-LA-101-04	6.02	3.18	2006-02-24	5.78	5.48	0.030	0.030	>50	2026-02-12	
4-LA-101-05	6.02	3.18	2006-02-24	5.81	5.54	0.024	0.027	>50	2026-02-12	
4-LA-101-06	6.02	3.18	2006-02-24	5.64	5.30	0.038	0.034	>50	2026-02-12	
4-LA-101-07	6.02	3.18	2006-02-24	5.57	4.87	0.056	0.070	24.1	2026-02-12	
4-LA-101-09	6.02	3.18	—	—	5.00	—	—	—	2026-02-12	

Line 4-LA-102 | Lean amine supply | A106 Gr.B | t_nom 6.02 mm | t-min 3.18 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
4-LA-102-01	6.02	3.18	2006-02-24	5.63	5.25	0.039	0.038	>50	2026-02-12	
4-LA-102-02	6.02	3.18	2006-02-24	5.67	5.36	0.036	0.031	>50	2026-02-12	
4-LA-102-03	6.02	3.18	2006-02-24	5.55	5.28	0.040	0.027	>50	2026-02-12	
4-LA-102-04	6.02	3.18	2006-02-24	5.34	4.90	0.055	0.044	31.5	2026-02-12	
4-LA-102-05	6.02	3.18	2006-02-24	5.62	5.34	0.033	0.028	>50	2026-02-12	
4-LA-102-06	6.02	3.18	2006-02-24	5.83	5.60	0.020	0.023	>50	2026-02-12	
4-LA-102-07	6.02	3.18	2006-02-24	5.63	5.40	0.030	0.023	>50	2026-02-12	
4-LA-102-08	6.02	3.18	—	—	5.70	—	—	—	2026-02-12	

Line 6-LA-103 | Lean amine header | A106 Gr.B | t_nom 7.11 mm | t-min 3.56 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
6-LA-103-01	7.11	3.56	2006-02-24	6.75	6.44	0.029	0.031	>50	2026-02-12	
6-LA-103-02	7.11	3.56	2006-02-24	6.87	6.53	0.033	0.034	>50	2026-02-12	
6-LA-103-03	7.11	3.56	2006-02-24	6.65	6.28	0.040	0.037	>50	2026-02-12	
6-LA-103-04	7.11	3.56	2006-02-24	6.52	6.15	0.051	0.037	>50	2026-02-12	
6-LA-103-05	7.11	3.56	2006-02-24	6.49	5.98	0.061	0.051	39.9	2026-02-12	
6-LA-103-06	7.11	3.56	2006-02-24	6.42	5.82	0.067	0.060	33.6	2026-02-12	
6-LA-103-07	7.11	3.56	2006-02-24	6.76	6.12	0.053	0.064	39.9	2026-02-12	
6-LA-103-08	7.11	3.56	2006-02-24	6.79	6.49	0.028	0.030	>50	2026-02-12	
6-LA-103-10	7.11	3.56	—	—	6.30	—	—	—	2026-02-12	

Line 4-RA-101 | Rich amine | A106 Gr.B | t_nom 8.56 mm | t-min 3.81 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
4-RA-101-01	8.56	3.81	2014-09-19	8.07	7.83	0.035	0.098	40.9	2019-02-12	
4-RA-101-02	8.56	3.81	2014-09-19	8.02	7.94	0.030	0.052	>50	2019-02-12	
4-RA-101-03	8.56	3.81	2014-09-19	7.88	7.65	0.048	0.066	>50	2019-02-12	
4-RA-101-04	8.56	3.81	2014-09-19	8.18	8.00	0.032	0.042	>50	2019-02-12	
4-RA-101-05	8.56	3.81	2014-09-19	7.98	7.82	0.039	0.068	>50	2019-02-12	
4-RA-101-06	8.56	3.81	2014-09-19	7.91	7.87	0.029	0.029	>50	2019-02-12	

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
4-RA-101-07	8.56	3.81	2014-09-19	8.22	8.14	0.023	0.032	>50	2019-02-12	
4-RA-101-08	8.56	3.81	2014-09-19	7.71	7.31	0.069	0.142	24.6	2019-02-12	
4-RA-101-09	8.56	3.81	2014-09-19	8.27	8.13	0.022	0.038	>50	2019-02-12	
4-RA-101-10	8.56	3.81	2014-09-19	8.21	8.02	0.033	0.052	>50	2019-02-12	
4-RA-101-11	8.56	3.81	2014-09-19	7.88	7.60	0.054	0.088	42.9	2019-02-12	

Line 4-RA-102 | Rich amine | A106 Gr.B | t_nom 8.56 mm | t-min 3.81 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
4-RA-102-01	8.56	3.81	2014-09-19	8.25	8.02	0.026	0.062	>50	2019-02-12	
4-RA-102-02	8.56	3.81	2014-09-19	7.89	7.64	0.047	0.070	>50	2019-02-12	
4-RA-102-03	8.56	3.81	2014-09-19	7.95	7.55	0.051	0.098	38.0	2019-02-12	
4-RA-102-04	8.56	3.81	2014-09-19	7.93	7.72	0.041	0.048	>50	2019-02-12	
4-RA-102-05	8.56	3.81	2014-09-19	7.84	7.67	0.044	0.080	48.1	2019-02-12	
4-RA-102-06	8.56	3.81	2014-09-19	7.95	7.73	0.039	0.056	>50	2019-02-12	
4-RA-102-07	8.56	3.81	2014-09-19	7.98	7.78	0.040	0.064	>50	2019-02-12	
4-RA-102-08	8.56	3.81	2014-09-19	7.58	7.35	0.057	0.106	33.3	2019-02-12	
4-RA-102-09	8.56	3.81	2014-09-19	7.84	7.54	0.051	0.088	42.2	2019-02-12	
4-RA-102-11	8.56	3.81	2014-09-19	8.12	7.96	0.046	0.046	>50	2019-02-12	

Line 6-RA-103 | Rich amine header | A106 Gr.B | t_nom 10.97 mm | t-min 4.57 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
6-RA-103-01	10.97	4.57	2014-09-19	10.62	10.29	0.036	0.036	>50	2019-02-12	
6-RA-103-02	10.97	4.57	2014-09-19	10.06	9.69	0.070	0.108	47.2	2019-02-12	
6-RA-103-03	10.97	4.57	2014-09-19	10.39	10.06	0.047	0.072	>50	2019-02-12	
6-RA-103-04	10.97	4.57	2014-09-19	9.79	9.31	0.084	0.132	35.8	2019-02-12	
6-RA-103-05	10.97	4.57	2014-09-19	9.81	9.48	0.074	0.124	39.5	2019-02-12	
6-RA-103-06	10.97	4.57	2014-09-19	10.43	10.32	0.035	0.048	>50	2019-02-12	
6-RA-103-07	10.97	4.57	2014-09-19	10.37	10.20	0.040	0.070	>50	2019-02-12	
6-RA-103-08	10.97	4.57	2014-09-19	10.49	10.33	0.032	0.022	>50	2019-02-12	
6-RA-103-09	10.97	4.57	2014-09-19	10.39	10.33	0.034	0.046	>50	2019-02-12	
6-RA-103-10	10.97	4.57	2014-09-19	9.93	9.49	0.077	0.112	43.8	2019-02-12	
6-RA-103-11	10.97	4.57	2014-09-19	10.10	9.77	0.236	0.236	22.0	2019-02-12	
6-RA-103-13	10.97	4.57	2014-09-19	10.38	10.15	0.060	0.060	>50	2019-02-12	

Line 3-RV-101 | Regenerator overhead | A106 Gr.B | t_nom 7.62 mm | t-min 3.56 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
3-RV-101-01	7.62	3.56	2013-09-09	6.49	6.34	0.063	0.081	34.2	2019-02-12	
3-RV-101-02	7.62	3.56	2013-09-09	6.44	6.31	0.063	0.052	43.5	2019-02-12	
3-RV-101-03	7.62	3.56	2013-09-09	5.80	5.63	0.095	0.098	21.2	2019-02-12	
3-RV-101-04	7.62	3.56	2013-09-09	6.13	5.88	0.086	0.102	22.8	2019-02-12	
3-RV-101-05	7.62	3.56	2013-09-09	6.48	6.36	0.066	0.050	42.6	2019-02-12	
3-RV-101-06	7.62	3.56	2013-09-09	6.11	5.96	0.082	0.087	27.7	2019-02-12	
3-RV-101-07	7.62	3.56	2013-09-09	5.73	5.54	0.122	0.122	16.2	2019-02-12	
3-RV-101-08	7.62	3.56	2013-09-09	6.36	6.31	0.074	0.074	37.2	2019-02-12	

Line 2-RV-102 | Regenerator overhead | A106 Gr.B | t_nom 5.54 mm | t-min 2.79 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
2-RV-102-01	5.54	2.79	2013-09-09	4.49	4.35	0.059	0.037	26.4	2019-02-12	
2-RV-102-02	5.54	2.79	2013-09-09	4.29	4.01	0.079	0.109	11.2	2019-02-12	
2-RV-102-03	5.54	2.79	2013-09-09	4.68	4.64	0.042	0.037	43.9	2019-02-12	

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
2-RV-102-04	5.54	2.79	2013-09-09	4.81	4.73	0.043	0.020	45.5	2019-02-12	
2-RV-102-05	5.54	2.79	2013-09-09	4.96	4.97	0.013	0.013	>50	2019-02-12	
2-RV-102-06	5.54	2.79	2013-09-09	4.55	4.27	0.102	0.102	14.6	2019-02-12	

Line 3-RF-101 | Reflux | A312 TP316 | t_nom 5.49 mm | t-min 2.41 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
3-RF-101-01	5.49	2.41	2013-12-09	5.37	5.31	0.009	0.017	>50	2026-02-12	
3-RF-101-02	5.49	2.41	2013-12-09	5.40	5.34	0.006	0.003	>50	2026-02-12	
3-RF-101-03	5.49	2.41	2013-12-09	5.37	5.33	0.006	0.012	>50	2026-02-12	
3-RF-101-04	5.49	2.41	2013-12-09	5.41	5.35	0.006	-0.002	>50	2026-02-12	
3-RF-101-05	5.49	2.41	2013-12-09	5.32	5.33	0.006	0.007	>50	2026-02-12	
3-RF-101-06	5.49	2.41	2013-12-09	5.17	5.23	0.015	0.017	>50	2026-02-12	
3-RF-101-07	5.49	2.41	2013-12-09	5.48	5.37	0.003	-0.002	>50	2026-02-12	
3-RF-101-09	5.49	2.41	2013-12-09	5.26	5.19	-0.029	-0.029	>50	2026-02-12	

Line 2-RF-102 | Reflux | A312 TP316 | t_nom 3.91 mm | t-min 2.16 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
2-RF-102-01	3.91	2.16	2013-12-09	3.60	3.76	0.008	0.009	>50	2026-02-12	
2-RF-102-02	3.91	2.16	2013-12-09	3.70	3.64	0.012	0.024	>50	2026-02-12	
2-RF-102-03	3.91	2.16	2013-12-09	3.79	3.75	0.009	0.005	>50	2026-02-12	
2-RF-102-04	3.91	2.16	2013-12-09	3.82	3.38	0.025	0.043	28.3	2026-02-12	
2-RF-102-05	3.91	2.16	2013-12-09	3.67	3.49	0.082	0.082	16.1	2024-03-06	
2-RF-102-06	3.91	2.16	2013-12-09	3.67	3.47	0.099	0.099	13.2	2022-09-26	
2-RF-102-07	3.91	2.16	2013-12-09	3.72	3.57	0.020	0.028	>50	2026-02-12	

Line 4-LA-104 | L/R exchanger, lean side | A106 Gr.B | t_nom 6.02 mm | t-min 3.18 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
4-LA-104-01	6.02	3.18	2006-02-24	5.75	5.31	0.038	0.044	48.2	2026-02-12	
4-LA-104-02	6.02	3.18	2006-02-24	5.56	5.00	0.052	0.056	32.4	2026-02-12	
4-LA-104-03	6.02	3.18	2006-02-24	5.74	5.42	0.027	0.032	>50	2026-02-12	
4-LA-104-04	6.02	3.18	2006-02-24	5.86	5.63	0.022	0.023	>50	2026-02-12	
4-LA-104-05	6.02	3.18	2006-02-24	5.51	4.88	0.058	0.063	26.9	2026-02-12	
4-LA-104-06	6.02	3.18	2006-02-24	5.83	5.55	0.028	0.028	>50	2026-02-12	
4-LA-104-07	6.02	3.18	2006-02-24	5.62	5.28	0.037	0.034	>50	2026-02-12	
4-LA-104-08	6.02	3.18	2006-02-24	5.50	5.11	0.042	0.039	46.3	2026-02-12	
4-LA-104-10	6.02	3.18	—	—	4.84	—	—	—	2026-02-12	

Line 4-RA-104 | L/R exchanger, rich side | A106 Gr.B | t_nom 8.56 mm | t-min 3.81 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
4-RA-104-01	8.56	3.81	2014-09-19	7.84	7.71	0.040	0.048	>50	2019-02-12	
4-RA-104-02	8.56	3.81	2014-09-19	8.09	7.85	0.035	0.058	>50	2019-02-12	
4-RA-104-03	8.56	3.81	2014-09-19	8.46	8.28	0.057	0.056	>50	2019-02-12	
4-RA-104-04	8.56	3.81	2014-09-19	8.49	8.18	0.063	0.084	>50	2019-02-12	
4-RA-104-05	8.56	3.81	2014-09-19	8.37	8.16	0.050	0.082	>50	2019-02-12	
4-RA-104-06	8.56	3.81	2014-09-19	7.91	7.61	0.049	0.074	>50	2019-02-12	
4-RA-104-07	8.56	3.81	2014-09-19	7.61	7.32	0.064	0.092	38.0	2019-02-12	
4-RA-104-08	8.56	3.81	2014-09-19	8.00	7.81	0.074	0.074	>50	2019-02-12	

Line 2-ST-101 | Reboiler steam | A106 Gr.B | t_nom 3.91 mm | t-min 2.29 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
2-ST-101-01	3.91	2.29	2006-02-24	3.64	3.61	0.016	0.003	>50	2026-02-12	
2-ST-101-02	3.91	2.29	2006-02-24	3.67	3.55	0.018	0.012	>50	2026-02-12	
2-ST-101-03	3.91	2.29	2006-02-24	3.82	3.55	0.020	0.027	46.5	2026-02-12	
2-ST-101-04	3.91	2.29	2006-02-24	3.77	3.68	0.007	0.009	>50	2026-02-12	
2-ST-101-05	3.91	2.29	2006-02-24	3.72	3.50	0.021	0.022	>50	2026-02-12	
2-ST-101-06	3.91	2.29	—	—	3.63	—	—	—	2026-02-12	

Line 3-CW-101 | Amine cooler CW | A106 Gr.B | t_nom 5.49 mm | t-min 2.79 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
3-CW-101-01	5.49	2.79	2006-02-24	5.13	5.03	0.020	0.010	>50	2026-02-12	
3-CW-101-02	5.49	2.79	2006-02-24	5.20	5.01	0.020	0.019	>50	2026-02-12	
3-CW-101-03	5.49	2.79	2006-02-24	5.08	4.93	0.030	0.015	>50	2026-02-12	
3-CW-101-04	5.49	2.79	2006-02-24	5.38	5.26	0.010	0.012	>50	2026-02-12	
3-CW-101-05	5.49	2.79	2006-02-24	5.33	5.35	0.007	-0.002	>50	2026-02-12	
3-CW-101-06	5.49	2.79	—	—	5.08	—	—	—	2026-02-12	

Line 3-CW-102 | Amine cooler CW | A106 Gr.B | t_nom 5.49 mm | t-min 2.79 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
3-CW-102-01	5.49	2.79	2006-02-24	5.20	5.06	0.022	0.014	>50	2026-02-12	
3-CW-102-02	5.49	2.79	2006-02-24	5.02	4.37	0.057	0.065	24.2	2026-02-12	
3-CW-102-03	5.49	2.79	2006-02-24	5.07	4.54	0.050	0.053	32.9	2026-02-12	
3-CW-102-04	5.49	2.79	2006-02-24	5.03	4.69	0.041	0.034	46.2	2026-02-12	
3-CW-102-05	5.49	2.79	—	—	4.66	—	—	—	2026-02-12	

Sign-off	Name	Function	Date	Signature
Prepared	Marcus Chen	lead UT technician, SNT-TC-1A UT/MT/PT Level II; CAN/CGSB-48.9712 UT Level II	2016-02-12	_____